CHIEF COMPLAINT: Oligomenorrhea, hirsutism, and adult acne with difficulty conceiving over 14 months.

HISTORY: The patient is a 26-year-old nulligravida who presents for evaluation of irregular menses. Menarche occurred at age 12, after which she experienced unpredictable cycles occurring every 60 to 90 days. For the past two years, she has had only four menses per year. She reports worsening coarse terminal hair growth over her chin, upper lip, and lower abdomen requiring frequent tweezing, as well as cystic facial acne. She and her partner have engaged in unprotected intercourse for 14 months without conception. She has gained 12 kg over the past two years.

PHYSICAL EXAMINATION:

General: Alert, well-nourished female with central adiposity. BMI is 31.8 kg/m2 (Weight 86 kg, Height 164 cm), BP 124/78 mmHg.

Skin and Hair: Modified Ferriman-Gallwey hirsutism score is 12 (abnormal > 8) with terminal dark hair over the chin, submental region, and linea alba. Moderate inflammatory papules and pustules over the jawline. Acanthosis nigricans noted along the posterior neck.

Pelvic Exam: Normal external female genitalia without clitoromegaly. Bimanual examination reveals a mobile, normal-sized uterus with mildly enlarged, non-tender bilateral adnexa.

LABORATORY INVESTIGATIONS:

Total Testosterone: Elevated at 78 ng/dL (reference 15-45 ng/dL); Free Testosterone elevated at 9.4 pg/mL.

LH/FSH Ratio: LH 14.2 mIU/mL, FSH 4.8 mIU/mL (LH:FSH ratio approximately 3:1).

Metabolic Profile: Fasting insulin elevated at 22 uIU/mL with fasting glucose 104 mg/dL (HOMA-IR 5.6, indicating insulin resistance). Serum DHEA-S normal at 180 ug/dL; 17-hydroxyprogesterone normal at 75 ng/dL (excludes non-classical congenital adrenal hyperplasia); TSH 1.8 uIU/mL; Prolactin normal at 12 ng/mL.

Pelvic Ultrasonography: Demonstrates normal uterine contour with 8 mm endometrium. Both ovaries are enlarged (right ovarian volume 13.2 mL, left ovarian volume 12.4 mL) with >20 small subcentimeter peripherally arranged follicles surrounding hyperechoic stroma ('string of pearls' appearance), fulfilling Rotterdam criteria for polycystic ovaries.

ASSESSMENT AND RECOMMENDATIONS:

1. Polycystic Ovary Syndrome (PCOS) fulfilling Rotterdam criteria (oligo-ovulation, hyperandrogenism, polycystic ovarian morphology).

2. Lifestyle and Metabolic Management: Target 5-10% weight loss with a low-glycemic Mediterranean dietary plan and regular resistance training to improve insulin sensitivity and restore ovulatory cyclicity.

3. Initiate oral metformin 500 mg daily, titrating to 1,500 mg daily.

4. Ovulation induction planning: Prescribe letrozole (2.5 mg daily on cycle days 3-7) following withdrawal bleed induced by oral medroxyprogesterone acetate.